CAPSTONE PROJECT – 2

Ratio Analysis of Listed Companies

A Comparative Study of BMW AG and Mercedes-Benz Group AG

Companies Studied: BMW AG | Mercedes-Benz Group AG

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COMPANY INTRODUCTION

BMW AG

- BMW AG (Bayerische Motoren Werke) is one of the world's leading premium automotive and motorcycle manufacturers, competing globally in luxury mobility.
- Headquartered in Munich, Germany, BMW was founded in 1916 and operates under four core brands — BMW, MINI, Rolls-Royce, and BMW Motorrad.
- The Quandt family (Susanne Klatten and Stefan Quandt) holds a combined promoter stake of roughly 50%, with the remainder held in public free float.
- The company is listed on the Frankfurt Stock Exchange (XETRA). For FY2024 it reported Group revenue of approximately €142,380 million and total assets of approximately €267,732 million.

BMW — Snapshot

Founded	1916 (Munich, Germany)
Headquarters	Munich, Bavaria
Sector	Premium Automobiles & Motorcycles
Listed On	Frankfurt Stock Exchange (XETRA)
Revenue (FY24)	≈ €142,380 million
Net Profit (FY24)	≈ €7,290 million

Mercedes-Benz Group AG

- Mercedes-Benz Group AG is one of the world's most iconic luxury automotive manufacturers, renowned for premium passenger cars and vans under the three-pointed star.
- Headquartered in Stuttgart, Germany, the company traces its lineage to 1926 and today operates the Mercedes-Benz Cars and Mercedes-Benz Vans divisions.
- Ownership is broadly held in public free float, with large long-term stakes held by investors including BAIC Group and Kuwait Investment Authority; there is no single controlling promoter.
- The company is listed on the Frankfurt Stock Exchange (XETRA). For FY2024 it reported Group revenue of approximately €145,594 million and total assets of approximately €265,010 million.

Mercedes-Benz — Snapshot

Founded	1926 (lineage to 1886)
Headquarters	Stuttgart, Baden-Württemberg
Sector	Luxury Automobiles & Vans
Listed On	Frankfurt Stock Exchange (XETRA)
Revenue (FY24)	≈ €145,594 million
Net Profit (FY24)	≈ €10,207 million

PRODUCT PORTFOLIO

Product Portfolio: BMW AG

1

Core Automotive

BMW 1–8 Series, X-Series SUVs, M Performance models

2

Electric Mobility (BEV)

BMW i4, i5, i7, iX, iX1, iX3 — the "Neue Klasse" line-up

3

Premium Sub-Brands

MINI (compact premium) and Rolls-Royce (ultra-luxury)

4

Motorcycles

BMW Motorrad — touring, adventure and roadster ranges

Product Portfolio: Mercedes-Benz Group AG

1

Core & Luxury Cars

A/C/E/S-Class, GLA–GLS SUVs, Maybach, AMG performance

2

Electric Mobility (EQ)

EQA, EQB, EQE, EQS — battery-electric passenger range

3

Vans Division

Sprinter, Vito, Citan — commercial and passenger vans

4

Mobility & Financial Services

Leasing, financing and connected-mobility solutions

COMPETITIVE LANDSCAPE

Competitors of BMW & Mercedes-Benz

- BMW's key competitors include Mercedes-Benz, Audi (Volkswagen Group), Porsche, Lexus (Toyota) and Tesla in the premium and EV segments.
- Mercedes-Benz's key competitors include BMW, Audi, Porsche, Lexus and increasingly Chinese premium entrants such as NIO and BYD's premium marques.
- Both companies compete directly in overlapping categories such as executive sedans, luxury SUVs and battery-electric premium vehicles.
- Intense category competition pushes both firms to continuously invest in electrification, autonomous-driving technology, software platforms and China market share.

BMW competes on:

Driving-dynamics engineering, brand heritage and premium pricing power across ICE and EV line-ups

Mercedes-Benz competes on:

Luxury comfort, ultra-premium S-Class/Maybach positioning and a broad commercial-vans franchise

COMPANY PROFILE

Key Additional Information

	BMW AG	Mercedes-Benz Group AG
Founded	1916	1926 (lineage to 1886)
Headquarters	Munich, Germany	Stuttgart, Germany
Ownership	Quandt family (~50%)	Broad public free float
Market Capitalisation*	≈ €48,000 million	≈ €51,000 million
Key Brands	BMW, MINI, Rolls-Royce	Mercedes-Benz, AMG, Maybach
Recent Development	"Neue Klasse" EV platform launch (2025–26)	Continued EQ electric line-up expansion

**Approximate market capitalisation, mid-2026.*

Liquidity Ratios

Current Ratio & Quick Ratio — BMW vs Mercedes-Benz

✓ 1. Current Ratio

Current Ratio = Current Assets ÷ Current Liabilities

BMW AG

= €96,387M ÷ €87,689M

= 1.10 : 1

Mercedes-Benz Group AG

= €101,767M ÷ €74,826M

= 1.36 : 1

Interpretation: Ideal range is 1.5:1 to 2:1. Both automakers sit below the ideal band; Mercedes-Benz (1.36) manages short-term obligations more comfortably than BMW (1.10), aided by a leaner current-liability base.

✓ 2. Quick Ratio

Quick Ratio = (Current Assets − Inventory) ÷ Current Liabilities

BMW AG

= (€96,387 − €23,291)M ÷ €87,689M

= 0.83 : 1

Mercedes-Benz Group AG

= (€101,767 − €26,234)M ÷ €74,826M

= 1.01 : 1

Interpretation: Ideal quick ratio is 1:1. Mercedes-Benz's 1.01 sits right at the ideal mark, while BMW's 0.83 reflects a larger inventory and finance-receivables-heavy current asset base typical of a captive-financing model.

Profitability Ratios (I)

Net Profit Ratio & Operating Profit Ratio

✓ 1. Net Profit Ratio

Net Profit Ratio = Net Profit ÷ Sales × 100

BMW AG

= €7,290M ÷ €142,380M × 100

= 5.12%

Mercedes-Benz Group AG

= €10,207M ÷ €145,594M × 100

= 7.01%

Interpretation: Ideal range is 10%-15%; both firms sit below this band, reflecting the capital-intensive, margin-compressed nature of premium autos in FY2024. Mercedes-Benz converts more of each euro of sales into profit than BMW.

✓ 2. Operating Profit Ratio

Operating Profit Ratio = Operating Profit (EBIT) ÷ Sales × 100

BMW AG

= €11,319M ÷ €142,380M × 100

= 7.95%

Mercedes-Benz Group AG

= €13,599M ÷ €145,594M × 100

= 9.34%

Interpretation: Ideal range is 15%-20%; both companies fall short in FY2024 amid tariff and cost headwinds. Mercedes-Benz's tighter operating cost control gives it roughly a 1.4-point margin edge over BMW.

Profitability Ratios (II)

Gross Profit Ratio & Return on Assets

✓ 1. Gross Profit Ratio

Gross Profit Ratio = Gross Profit ÷ Sales × 100

BMW AG

= €20,980M ÷ €142,380M × 100

= 14.73%

Mercedes-Benz Group AG

= €28,576M ÷ €145,594M × 100

= 19.63%

Interpretation: Mercedes-Benz's higher gross margin (19.63% vs 14.73%) reflects stronger pricing power in its S-Class/Maybach/AMG mix, while BMW's broader volume segments carry relatively thinner gross spreads.

✓ 2. Return on Assets (ROA)

Return on Assets = Net Profit ÷ Total Assets × 100

BMW AG

= €7,290M ÷ €267,732M × 100

= 2.72%

Mercedes-Benz Group AG

= €10,207M ÷ €265,010M × 100

= 3.85%

Interpretation: Ideal ROA is above 15%-20% for typical firms, but both automakers' balance sheets are inflated by captive-financing receivables, depressing this ratio. Mercedes-Benz still deploys assets more efficiently than BMW.

Return Ratios

Return on Equity & Return on Capital Employed

✓ 1. Return on Equity (ROE)

Return on Equity = Net Profit ÷ Shareholders' Equity × 100

BMW AG

= €7,290M ÷ €95,003M × 100

= 7.67%

Mercedes-Benz Group AG

= €10,207M ÷ €93,630M × 100

= 10.90%

Interpretation: Ideal ROE is 15%-20%; both automakers fall below this in a tariff-affected FY2024. Mercedes-Benz's higher net margin lifts its ROE roughly 3 points above BMW's.

✓ 2. Return on Capital Employed (ROCE)

ROCE = EBIT ÷ Capital Employed × 100

BMW AG

= €11,319M ÷ €180,043M × 100

= 6.29%

Mercedes-Benz Group AG

= €13,599M ÷ €190,184M × 100

= 7.15%

Interpretation: Ideal ROCE is above 15%. Both are below this benchmark, but Mercedes-Benz's 7.15% edges out BMW's 6.29%, showing marginally more efficient use of total capital to generate operating returns.

Solvency (Leverage) Ratios

Debt-Equity Ratio & Interest Coverage Ratio

✓ 1. Debt-Equity Ratio

Debt-Equity Ratio = Total Debt ÷ Shareholders' Equity

BMW AG

= €111,261M ÷ €95,003M

= 1.17 : 1

Mercedes-Benz Group AG

= €112,798M ÷ €93,630M

= 1.20 : 1

Interpretation: Ideal D/E is below 1. Both firms exceed this, largely because reported debt includes their captive financial-services borrowings used to fund customer leasing and financing books, not just operating debt.

✓ 2. Interest Coverage Ratio

Interest Coverage Ratio = EBIT ÷ Interest Expense

BMW AG

= €11,319M ÷ €542M

= 20.88 times

Mercedes-Benz Group AG

= €13,599M ÷ €190M

= 71.57 times

Interpretation: Ideal ICR is above 3-4 times. Both comfortably cover interest obligations; Mercedes-Benz's far lower net interest expense gives it a much larger cushion (71.57x) than BMW (20.88x).

Solvency & Efficiency Ratios

Proprietary Ratio & Inventory Turnover Ratio

✓ 1. Proprietary Ratio

Proprietary Ratio = Shareholders' Equity ÷ Total Assets

BMW AG

= €95,003M ÷ €267,732M

= 0.35 (35%)

Mercedes-Benz Group AG

= €93,630M ÷ €265,010M

= 0.35 (35%)

Interpretation: Ideal range is 0.5-0.67. Both companies fall below the ideal band and are nearly identical, financing roughly a third of assets through owners' funds — consistent with capital-intensive automakers running large captive-finance books.

✓ 2. Inventory Turnover Ratio

Inventory Turnover Ratio = Cost of Goods Sold ÷ Inventory

BMW AG

= €121,400M ÷ €23,291M

= 5.21 times

Mercedes-Benz Group AG

= €117,018M ÷ €26,234M

= 4.46 times

Interpretation: A higher ratio indicates faster stock movement. BMW turns inventory over somewhat faster (5.21x) than Mercedes-Benz (4.46x), reflecting a leaner vehicle-inventory cycle relative to sales.

Efficiency (Turnover) Ratios

Asset, Receivables & Fixed Asset Turnover Ratios

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Asset Turnover Ratio = Sales ÷ Total Assets

BMW: €142,380M ÷ €267,732M = 0.53x

Mercedes-Benz: €145,594M ÷ €265,010M = 0.55x

Mercedes-Benz generates €0.55 of sales per €1 of assets vs BMW's €0.53 — a marginal edge in overall asset deployment.

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Receivables Turnover Ratio = Sales ÷ Trade Receivables

BMW: €142,380M ÷ €6,309M = 22.57x

Mercedes-Benz: €145,594M ÷ €46,293M = 3.15x

BMW collects trade dues far faster (22.57x). Mercedes-Benz's reported receivables include a large captive-financing/leasing book, which inflates the balance and depresses this ratio — the two are not directly comparable without adjustment.

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Fixed Asset Turnover Ratio = Sales ÷ Fixed Assets

BMW: €142,380M ÷ €171,345M = 0.83x

Mercedes-Benz: €145,594M ÷ €163,243M = 0.89x

Mercedes-Benz generates slightly more sales per euro of fixed assets (0.89x vs 0.83x) — a marginally leaner non-current asset base relative to revenue.

CONCLUSION

Conclusion & Overall Comparison

Profitability & Margins	Mercedes-Benz leads on Net Profit, Operating and Gross Profit ratios — reflecting stronger pricing power in its luxury sedan and AMG mix.
Returns to Investors	Mercedes-Benz's ROE (10.90%) and ROA are higher; it also edges ahead on ROCE, showing more efficient use of total capital employed.
Liquidity	Mercedes-Benz maintains healthier Current (1.36) and Quick (1.01) ratios, giving it a stronger short-term cash cushion than BMW.
Solvency	Both firms carry D/E above 1, largely from captive-financing debt; Mercedes-Benz's far higher interest coverage (71.57x vs 20.88x) reflects lower net interest cost.
Efficiency	BMW turns over inventory and collects receivables faster; Mercedes-Benz shows marginally better asset and fixed-asset utilisation.

Overall Verdict

Mercedes-Benz Group is the stronger performer on profitability margins, liquidity and return on equity in FY2024, driven by its premium luxury mix and disciplined cost control. BMW demonstrates faster receivables collection and inventory turnover, and a comparable proprietary ratio. Together, both companies represent financially resilient but strategically distinct premium automotive business models navigating a challenging tariff and EV-transition environment.